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Genomics

Horizontal acquisition of the Type I restriction–modification system enhances bacterial pathogenicity by mediating methylation of transcription factor-encoding genes

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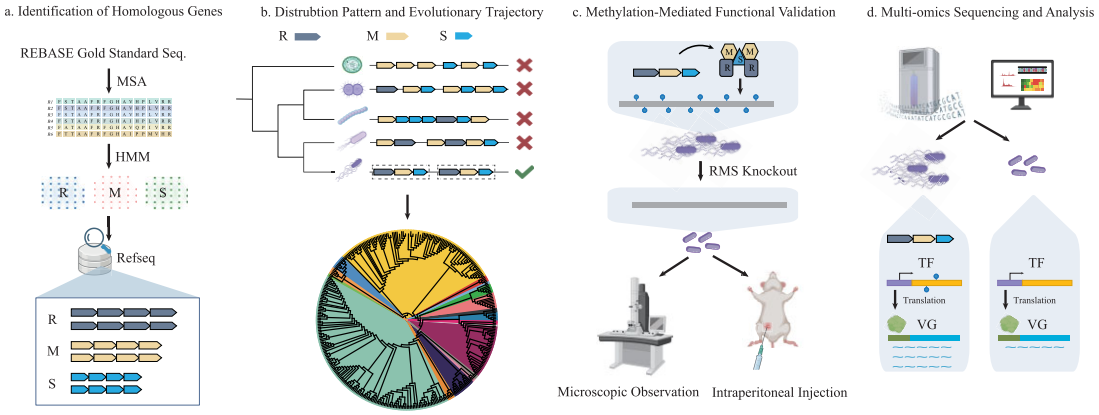
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Abstract

The Type I restriction–modification (RM) system, encoded by the *hsdR*, *hsdM*, and *hsdS* genes, plays a crucial role in shaping the prokaryotic DNA methylation landscape. Although known for defending against foreign DNA, key aspects of its evolutionary trajectory and functional implications after stable inheritance remain poorly understood. In this study, we identified four primary types of Type I RM systems across 4273 prokaryotic genomes based on gene arrangement. Among these, the 5′-*hsdR*, *hsdM*, *hsdS*-3′ (RMS) configuration emerged as the most evolutionarily advanced form. Phylogenetic reconstruction revealed that RMS was formed through gene duplication, horizontal gene transfer, and gene loss, and it now stably exists in bacteria. Functional characterization demonstrated that RMS deletion in bacteria led to the absence of flagella and a significant reduction in their ability to colonize and infect mice. Integrated multi-omics analysis uncovered a potential regulatory cascade where RMS modulates the expression of transcription factors via DNA methylation, which in turn regulate downstream flagellar and chemotaxis genes, thereby influencing bacterial pathogenicity. These findings establish a complete evolutionary–functional paradigm, elucidating how (evolutionary trajectory) and why (functional constraints) RMS has been stably inherited in bacterial genomes, and revealing the molecular mechanism through which RMS orchestrates bacterial pathogenicity.

Graphical abstract



Introduction

DNA methylation is a common epigenome modification in prokaryotes. In a survey of the genomes of 230 prokaryotes, over 90% were found to contain methylated sites [1]. DNA methylation occurs on the bases of cytosine (C) and adenine (A) to form *N*4-methylcytosine (4mC), *C*5-methylcytosine (5mC), and *N*6-methyladenine (6mA) in prokaryotes [2]. These modifications not only protect the DNA from cleavage by endonuclease, but also play key roles in nucleoid organization, cell cycle control, DNA repair, and regulation of

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transcription [3]. For many years, DNA methylation was thought to predominantly occur in promoter regions, where it acted as a repressive epigenetic mark to downregulate gene expression. However, more recent studies have revealed that DNA methylation is more prevalent within gene bodies than in promoters. In particular, intragenic methylation may play an active role in modulating gene expression [4–6].

The restriction–modification (RM) system, referred to as the innate immune system in prokaryotes, protects the host's DNA by methylating its own sequences, while restricting the invasion of unmethylated foreign DNA such as phages [7]. RM systems are classified into four types (I–IV) based on their subunit composition, cofactor requirement, sequence recognition, and cleavage pattern, with the Type I RM system being the first to be discovered [8, 9]. The Type I RM system is encoded by three *hsd* genes: *hsdR*, *hsdM*, and *hsdS*. HsdS specifically recognizes the DNA sequence, HsdM binds *S*-adenosylmethionine (AdoMet) to catalyze methylation of bases, and HsdR translocates DNA in an ATP-dependent process and cleaves unmethylated DNA [10]. In canonical Type I RM system, *hsdM* and *hsdS* share a common promoter, while *hsdR* is driven by a separate promoter. These genes are typically arranged in the genome in the order of 5′-*hsdR*, *hsdM*, *hsdS*-3′ (RMS) [8]. However, recent studies have identified alternative arrangement of Type I RM systems in certain species. For instance, in *Pseudomonas aeruginosa*, the genes are organized in the order of 5′-*hsdR*, *hsdS*, *hsdM*-3′ (RSM) [11]. Additionally, 5′-*hsdM*, *hsdS*-3′ (MS) can function in conjunction with *hsdR*, even when located at distant regions of the genome [12]. Despite these insights, the distribution and diversity of Type I RM systems across prokaryotes remain unexplored.

Horizontal gene transfer (HGT) is a key process in microbial evolution and ecology. It enables the rapid spread of genetic material across different species, contributing to genetic diversity, the acquisition of new traits, and adaptation to environmental changes [13]. Current evolutionary models propose that Type I RM systems first emerged in prokaryotes as primitive defense mechanisms during early host-phage coevolution [14]. Subsequently, under conditions of low selection pressure, hosts may benefit from eliminating these energy-costly Type I RM systems via HGT-mediated genetic exchange, thereby reducing their metabolic burden. This phenomenon also highlights their role as a pan-immune system [15, 16]. Through HGT, neighboring strains can transiently acquire Type I RM systems, enabling rapid adaptation to diverse phage challenges. However, a comprehensive study on the origin and evolutionary history of Type I RM systems is still lacking. In addition to defending against foreign DNA invasion, recent studies have shown that the Type I RM system can also influence bacterial virulence. When the Type I RM system (RSM) of *P. aeruginosa* is knocked out, 92 genes are differentially expressed, and the ΔRSM strain exhibits a marginally but significantly reduction in virulence compared to the wild-type (WT) strain [11]. Similarly, in *Streptococcus pyogenes*, 20 genes are significantly downregulated in the ΔRSM strain relative to the WT strain. Mice infected with the ΔRSM strain exhibit an enhanced host immune response, characterized by larger skin lesions, and elevated levels of proinflammatory cytokines [17]. However, the regulatory mechanism remains unclear, particularly regarding the biological processes and mechanisms controlled by the canonical Type I RM system (RMS).

In this study, we systematically investigated (i) the phylogenetic distribution patterns of four distinct Type I RM systems across prokaryotes, identifying key ecological and geographical factors influencing their prevalence; (ii) the evolutionary trajectory of RMS, which may emerge as the genomically stabilized end product of successive gene duplication, HGT, and gene loss events; and (iii) the functional consequences of RMS maintenance, particularly its methylation-dependent regulation of bacterial pathogenicity following genomic fixation. Through these analyses, we bridge evolutionary dynamics with phenotypic outcomes, and provide novel insights into the molecular mechanism by which RMS regulates bacterial pathogenicity.

Materials and methods

Identification of Type I RM systems and prophages

The protein sequences of *hsdR*, *hsdM*, and *hsdS* were extracted from the protein_gold_seqs file stored in the REBASE database [18]. Redundant sequences were removed using MMseq2 [19] with a 30% sequence identity threshold. Multiple sequence alignments (MSAs) were performed using MAFFT [20], and hidden Markov models (HMMs) were constructed using hmmbuild from HMMER [21]. Completely assembled reference genome sequences, protein sequences, and gene annotations for 4019 bacteria and 254 archaea were downloaded from the NCBI Reference Sequence (RefSeq) database using NCBI Datasets [22, 23]. The *hsdR*, *hsdM*, and *hsdS* were identified in these species by sequence alignments against the HMMs using hmmsearch from HMMER with a threshold of $e < 1\times10^{-10}$. The Type I RM system, arranged in the order of *hsdR*, *hsdM*, and *hsdS* on the genome without any intervening genes, was referred to as RMS. The Type I RM system in which *hsdR* and *hsdM* were separated by no more than five genes, while *hsdM* and *hsdS* were closely connected, was designed as R-MS. Similarly, RSM and R-SM were also defined. Completely assembled genome sequences, protein sequences, and gene annotations for 41 *Aeromonas veronii* strains were downloaded from the RefSeq database using NCBI Datasets [22, 23]. The *hsdR*, *hsdM*, *hsdS* genes, along with various configurations of Type I RM systems, were identified in these strains. Prophages were detected from genome sequences using VIBRANT [24]. We systematically enumerated all four configurations (RMS, R-MS, RSM, R-SM) of Type I RM systems within each bacterial and archaeal phylum. The frequency of each configuration was then calculated as the mean number of systems per species within each phylum, using the following formula: frequency = (total count of configuration X in phylum Y)/(number of species in phylum Y).

Construction of phylogenetic trees

A total of 120 bacterial and 53 archaeal marker genes were identified from 4019 bacterial and 254 archaeal genomes, respectively, and subsequently aligned and concatenated using GTDB-Tk [25]. The phylogenetic tree at the phylum level was constructed using the WAG model of amino acid evolution in FastTree [26]. Single-copy genes were extracted from *A. veronii* strains using OrthoFinder [27], and MSAs were performed using MUSCLE [28]. The alignment results were processed with TRIMAL [29] for automatic trimming using the automated1 parameter, and subsequently used to con-

struct the phylogenetic tree of *A. veronii* strains using IQ-TREE [30]. Species containing RMS, as well as the *hsdR*, *hsdM*, and *hsdS* genes that made up RMS, were extracted using a Perl script. Similarly, 120 bacterial and 53 archaeal marker genes were extracted from these species, and subsequently aligned and concatenated using GTDB-Tk [25]. Homologous sequences of HsdR, HsdM, and HsdS were aligned using MAFFT [20]. The best-fit models were determined, and phylogenetic trees were constructed for the species containing RMS as well as the *hsdR*, *hsdM*, and *hsdS* genes using IQ-TREE [30]. All phylogenetic trees were visualized using iTOL [31], and their reliabilities were evaluated using 1000 bootstraps.

Evolutionary inference of *hsdR, hsdM,* and *hsdS*

The core program OptRoot in RANGER-DTL 2.0 [32] was used to optimally root the phylogenetic trees of *hsdR*, *hsdM*, and *hsdS*. Each rooted gene tree was compared against the species tree to compute reconciliations using the core program Ranger-DTL. For each gene, 1000 reconciliations were generated with different random seeds, and subsequently input into the core program AggregateRanger to compute support values. A gene duplication or HGT event was considered supported only if the support value was >50%.

Construction of ΔRMS strain

The RMS knockout (ΔRMS) strain in *A. veronii* C4 was generated using the *sacB* homologous recombination method (Supplementary Fig. S1) [33]. The upstream and downstream fragments of *hsdR*, *hsdM*, and *hsdS* were amplified by polymerase chain reaction (PCR) using the *A. veronii* C4 genome as the template. The primers used are listed in Supplementary Table S1. The flanking regions were then ligated as fusion fragments by overlap-extension PCR and cloned into the suicide plasmid pRE112 [34] using the ClonExpress II One Step Cloning Kit (Vazyme, China). The recombinant plasmids with the correct sequences, named pRE112-Δ*hsd*R (containing the upstream and downstream homology arms of *hsdR*), pRE112-Δ*hsdM* (containing the upstream and downstream homology arms of *hsdM*), and pRE112-Δ*hsdS* (containing the upstream and downstream homology arms of *hsdS*), were transferred into *Escherichia coli* WM3064 competent cells via electroporation. *E. coli* WM3064 is a diaminopimelic acid (DAP)-auxotrophic strain (genotype: Δ*dapA::erm*) derived from *E. coli* β2155, which serves as an efficient conjugation donor for Gram-negative bacteria. pRE112-Δ*hsdR* was then conjugated into *A. veronii* C4 by conjugative transfer and selected on LB solid plates containing 22% sucrose and ampicillin. Positive clones were confirmed using specific primers (Supplementary Table S1). pRE112-Δ*hsdM* was conjugated into the Δ*hsdR* strain, and pRE112-Δ*hsdS* was conjugated into the Δ*hsdRM* strain, resulting in the ΔRMS strain.

Construction of ΔRMS/pRMS strain

The *hsdRMS* gene cluster was amplified by PCR using the *A. veronii* C4 genome as the template. The full-length *hsdRMS* DNA fragment was then inserted into the linearized pBBR vector (digested with KpnI and EcoRI) using the One Step CloneExpress kit (Vazyme, China). The resulting pBBR-*hsdRMS* (pRMS) plasmid was subsequently transformed into *E. coli* WM3064 competent cells via heat shock. The complemented strain ΔRMS/pRMS was generated by conjugating *A. veronii* ΔRMS with *E. coli* WM3064 carrying the pRMS plasmid. Selection was carried out on solid medium containing both ampicillin and kanamycin, and the ΔRMS/pRMS strain was confirmed by colony PCR.

Bacterial morphology observation

Single colonies of *A. veronii* C4 WT and ΔRMS strains were picked and cultured in LB liquid medium. After shaking incubation at 30°C for 12 h, the cultures were subcultured into fresh LB medium at a 1% inoculation rate. Following an additional 6 h of incubation, 10^7 cells were collected, centrifuged at 6000 rpm for 3 min, and washed twice with phosphate buffered saline. The bacterial pellets were then resuspended in 1 ml PBS, and 10 μl was placed on a copper grid and allowed to air dry. Flagellar morphology was subsequently observed under a transmission electron microscope (TEM, HT7700).

In vivo colonization and pathological assessment

Fifteen four-week-old male Kunming mice were purchased from Hainan Pharmacology Research Center and divided into three groups. One group served as a control, while the other two groups were intraperitoneally injected with *A. veronii* C4 WT and ΔRMS strains at a dose of 10^5 CFU/g of body weight, respectively. After 8 h of infection, the mice were euthanized by cervical dislocation. Hearts, livers, spleens, lungs, and kidneys were dissected and divided into two parts. One part was fixed with 4% paraformaldehyde for 24 h, then sent to Wuhan Servicebio Technology Co., Ltd for formalin fixation and paraffin embedding, followed by hematoxylin and eosin (H&E) staining. The samples were imaged using a microscope (Servicebio, China). The other part was homogenized with 1 ml of 1× PBS using a tissue grinder at 2000 rpm for 3 min. The supernatant was then plated onto agar plates containing ampicillin and incubated at 30°C for 24 h. Colony counts were determined, and differences were assessed using ANOVA. All the animal experiments were approved and conducted in compliance with the ethical guidelines and recommendations established by the ethical committee of the Hainan University (approval number HNUAUCC-2024-00082).

Transcriptome sequencing and analysis

Single colonies of *A. veronii* C4 WT and ΔRMS strains were picked and cultured in LB liquid medium at 30°C for 16 h. The cells were then collected and stored at −80°C for transcriptome and DNA methylation sequencing. Total RNA was extracted using the TRIzol reagent (Invitrogen, USA), and was subsequently analyzed and quantified using an Agilent 2100 Bioanalyzer (Agilent, USA). DNA probe was designed to be complementary to a specific region of ribosomal RNA. RNase H was used to digest the RNA strand of DNA/RNA hybrid, while the remaining DNA probe was digested by DNase I. The purified RNA was then fragmented. First-strand complementary DNA (cDNA) was generated using random hexamer-primed reverse transcription, followed by a second-strand cDNA synthesis. The synthesized double-stranded cDNA was subject to end repair, A-tailing, and adaptor ligation. The resulting library was amplified by PCR and subsequently sequenced on the Beijing Genomics Institute DNBSEQ platform. The sequencing data were filtered using SOAPnuke [35], with the following steps: (i) removal of reads containing adapter;

(ii) removal of reads with >20% low-quality bases (base quality ≤ 15); and (iii) removal of reads with >5% unknown base ("N" base). Afterward, clean reads were obtained and mapped to the reference genome using HISAT2 [36]. Gene expression levels were quantified using RSEM [37]. Differentially expressed genes (DEGs) were identified using DESeq2 [38] with a corrected *P*-value ≤ 0.05. All genes were annotated by sequence alignment against the Kyoto Encyclopedia of Genes and Genomes (KEGG) [39] database. KEGG enrichment analysis of DEGs was performed using the phyper function in R, with a corrected *P*-value ≤ 0.05.

DNA methylation sequencing and analysis

Genomic DNA was extracted from *A. veronii* C4 WT and ΔRMS strains using the FastPure Bacteria DNA Isolation Mini Kit (Vazyme, China). After fragmentation with the Covaris g-TUBE, the SMRTbell Prep Kit 3.0 (Pacific Biosciences) was used for library construction according to the manufacturer's instructions. The procedure involves the following steps: (i) repair and A-tailing: repairing sites of DNA damage and preparing the sheared DNA for ligation to the SMRTbell adapter; (ii) adapter ligation and cleanup: ligating the SMRTbell adapter to the ends of each DNA fragment; (iii) nuclease treatment: removing unligated DNA fragments and leftover SMRTbell adapter from the sample; and (iv) size selection: selecting fragments within the optimal 10–20 kb range. The final library was quality-checked using an Agilent 2100 Bioanalyzer (Agilent, USA) before being sequenced on the PacBio RSII platform at Wuhan IGENEBOOK Biotechnology Co., Ltd. The sequencing data were analyzed using the PacBio SMRT analysis platform (6.0.0) (https://www.pacb.com/products-and-services/analytical-software/smrt-analysis/), with the following steps: (i) data in h5 format were converted to bam format using bax2bam, and an index of the reference genome was built using Samtools; (ii) bam format data were filtered and aligned to the reference genome using Pbalign; (iii) 6mA sites with >25-fold coverage were identified using ipdSummary, and 20 bases from their upstream and downstream sequences were extracted; and (iv) motifs were identified using MotifMaker (https://github.com/wangdi2014/MotifMaker). The motif that differed between the WT and ΔRMS strains was the one recognized and methylated by RMS. TF-encoding genes containing the RMS target sites were extracted, and their regulatory sites were obtained from the RegPrecise database [40]. For each TF, sites2meme from MEME was used to convert its regulatory site files into a MEME motif [41]. Genome-wide motif scanning was performed against the *A. veronii* C4 genome using MAST [11]. Putative TF–target interactions were defined when motif matches were found within 150 bp upstream of transcription start sites.

RT-qPCR

Total RNA was extracted from *A. veronii* C4 WT and ΔRMS strains using the TRIzol reagent (Invitrogen, USA). Reverse transcription was performed in duplicate using the Prime Script RT Kit (Vazyme, China), followed by qPCR using the SYBR Premix Ex Taq II kit (Vazyme, China) on a Light Cycler System (Roche, USA). Primers for Reverse Transcription Quantitative PCR (RT-qPCR) were synthesized by Sangon Biotech Co., Ltd and are listed in Supplementary Table S2. Relative gene expression levels were calculated using the $2^{-\Delta\Delta Ct}$ method, with *gyrB* as the reference gene. The expression differences were assessed using ANOVA.

Results

Distribution of Type I RM systems in prokaryotes is diverse

The *hsdR*, *hsdM*, and *hsdS* genes were identified in prokaryotes, including 4019 bacterial and 254 archaeal species (Fig. 1A). The number and distribution of *hsdR*, *hsdM*, and *hsdS* genes varied considerably among these species (Supplementary Fig. S2). Specifically, 64.6%, 75.5%, and 60.2% of the species contained the *hsdR*, *hsdM*, and *hsdS* genes, respectively (Fig. 1A). On average, each species harbored 1.8 *hsdR* genes, 2.3 *hsdM* genes, and 2 *hsdS* genes (Fig. 1A), suggesting that these genes have undergone replication or HGT. Four main types of Type I RM systems were identified in these species: RMS, RSM, R-MS (where "-" indicates the presence of no more than five inserted genes between *hsdR* and *hsdMS*), and R-SM (Fig. 1B).

Compared to R-MS, which contained gene insertions, RMS was more prevalent in bacteria (Fig. 1C). Similarly, RSM was more common than R-SM in bacteria (Fig. 1C), while archaea showed opposite trends with R-MS/R-SM predominance (Fig. 1C). For a more detailed analysis, the frequencies of the four configurations of Type I RM systems were calculated across different phyla (Fig. 1D). Among bacteria, RMS was predominantly found in species of Fusobacteriota and Desulfobacteriota, with an average of over 0.2 RMS per species. R-MS was rarely detected in species from any phylum. RSM was most prevalent in species of Firmicutes, Actinobacteriota, Planctomycetota, and Myxococcota (>0.2 systems/species). R-SM was more commonly found in species of Campylobacterota, Planctomycetota, and Desulfobacterota (>0.2 systems/species) (Fig. 1D). In contrast, archaeal phyla uniformly exhibited low frequencies of all four configurations, with none surpassing 0.2 systems per species in any lineage (Fig. 1D).

Notably, despite the relatively high number of individual genes *hsdR*, *hsdM*, and *hsdS* in Cyanobacteria (Supplementary Fig. S3), none of the four configurations of Type I RM systems were commonly observed within this phylum (Fig. 1D). This suggested that, in Cyanobacteria, these genes tended to be scattered throughout the genome, rather than forming a gene cluster (Supplementary Fig. S4A). In contrast, both the number of individual genes and Type I RM systems were relatively high in Desulfobacterota (Supplementary Fig. S3 and Fig. 1D), with a diverse array of Type I RM systems, especially a higher number of R-SM compared to RSM (Supplementary Fig. S4B). Furthermore, we examined the frequency of prophages in species containing Type I RM systems and found a positive correlation between RMS and prophages (Fig. 1E). This suggests that prophages may act as carriers for RMS, facilitating its HGT among different species.

Distribution of Type I RM systems correlates with the ecological niche of strains

To further investigate the distribution patterns of Type I RM systems within species, *A. veronii* was chosen as the model organism, and 42 strains were analyzed (Fig. 2A). *A. veronii*

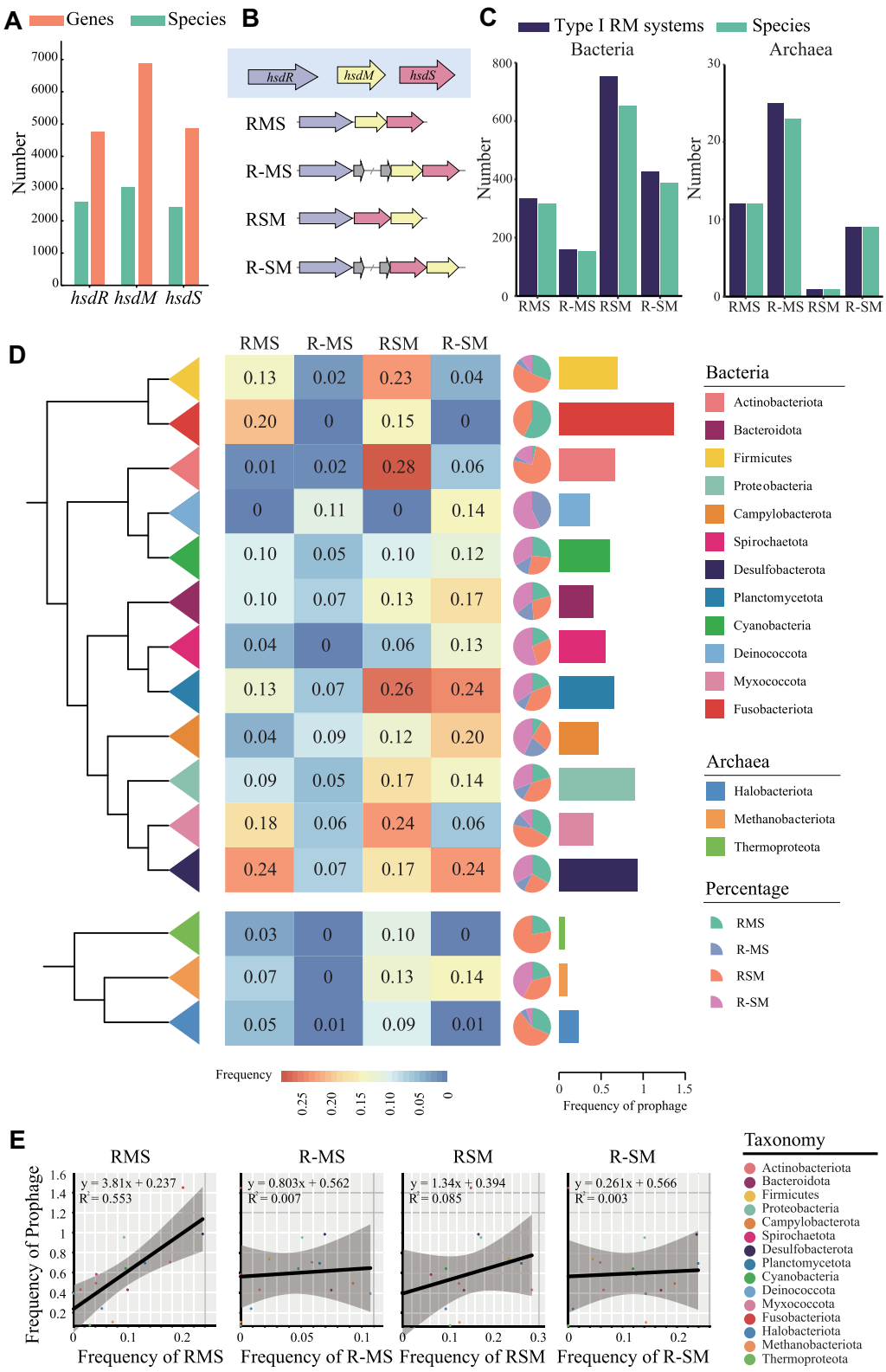


Figure 1. Distribution of Type I RM systems and prophages in prokaryotes. (**A**) Number of *hsd* genes and the number of species harboring the corresponding *hsd* genes. (**B**) Gene arrangements of the four configurations of Type I RM systems. (**C**) Number of Type I RM systems and the number of species harboring the corresponding Type I RM systems. (**D**) Frequencies of Type I RM systems across various phyla. The phylogenetic tree illustrates the evolutionary relationships among phyla. The heatmap (with a blue-to-red gradient) displays the frequency of Type I RM systems. The pie chart represents the proportion of each type of Type I RM systems. The bar chart shows the frequency of prophages. (**E**) Correlation between the frequencies of Type I RM systems and prophages at the phylum level.



Figure 2. Distribution of Type I RM systems and prophages in *A. veronii* strains. (**A**) Distribution of *hsd* genes in *A. veronii* strains. The phylogenetic tree illustrates the evolutionary relationships among strains, with geographic location and isolation source indicated by different colors. The gene cluster shows the relative positions of *hsd* genes within the genome. The bar chart represents the number of prophages in each strain. (**B**) Phylogenetic trees of *hsdR*, *hsdM*, and *hsdS* genes in *A. veronii*. Genes encoding the same type of Type I RM systems (RMS, R-MS, RSM, or R-SM) are highlighted with the same background color. Genes without background color do not encode any Type I RM system.

is a Gram-negative bacterium known for its strong motility, virulence, and environmental adaptability [42]. It is widely distributed in aquatic environments, and can infect aquatic organisms and humans, posing a significant threat to aquaculture and human health [43, 44]. As expected, not all *A. veronii* strains contained Type I RM systems, and the types of these systems varied significantly across strains, even within the same clade. For instance, none of the strains, isolated from Thailand and named starting with "colony", possessed Type I RM systems. In contrast, the C4 strain, which clustered with the colony512 strain, contained two types of Type I RM systems: RMS and R-SM (Fig. 2A). A similar pattern was observed in the MS-18-37 and FDAARGOS_632 strains. Notably, even strains from the same isolation source or geographic location exhibited considerable differences in the number and types of Type I RM systems. For instance, the WP3-W19-ESBL-03 strain contained seven *hsd* genes and only R-SM, while the WP9-W18-ESBL-04 strain, with more than twice the number of *hsd* genes, harbored both R-SM and RSM (Fig. 2A).

When comparing strains from different isolation sources, those isolated from water generally exhibited a higher number of Type I RM systems (Fig. 2A), with an average of more than two per strain. In contrast, strains isolated from humans contained fewer, with an average of only 0.8 per strain. Additionally, strains from water also harbored more prophages, with an average of 3.1 per strain (Fig. 2A). When comparing strains from different geographic locations, those isolated from China and Japan had more Type I RM systems, whereas strains from Thailand lacked them entirely (Fig. 2A). These findings suggest that the distribution of Type I RM systems is shaped by a combination of factors, including isolation source and geographic location.

Compared to the other two types of Type I RM systems, RMS and R-SM were more frequently found in *A. veronii* (Fig. 2A). The *hsdR*, *hsdM*, and *hsdS* genes, which constituted RMS, clustered within the same group of clades across all three gene trees (Fig. 2B). This suggests that RMS is genetically stable in *A. veronii*. Notably, the branch lengths of *hsdR* and *hsdM* were very short, indicating that the evolutionary distances of these genes among different strains are minimal. In contrast, the *hsd* genes associated with R-SM were dispersed across various clades, some of which contained genes that did not belong to any Type I RM system (Fig. 2B). These genes are likely remnants of Type I RM systems that were disrupted during HGT.

RMS has undergone a long and complex evolutionary process

We extracted the species harboring RMS from 4019 bacteria and 254 archaea. Phylogenetic trees for these species (Fig. 3A) as well as for the *hsdR*, *hsdM*, and *hsdS* genes (Fig. 3B) forming RMS were constructed. Notably, there were distinct topologies between the species tree and the gene trees (Fig. 3A and B), suggesting that these genes primarily evolve through HGT across prokaryotes. This conclusion was further supported by nucleotide composition analysis, which demonstrated significant deviations in the GC content between the horizontally transferred *hsd* genes and their recipient genomes (P-value = 0.01 for *hsdR*, P-value = 3.62×10^{-10} for *hsdM*, and P-value = 4.87×10^{-8} for *hsdS*; Wilcoxon signed-rank test; Supplementary Tables S3–S5). In contrast, these genes underwent relatively few duplications, with such events occurring only in three species within Firmicutes (Fig. 3C and Supplementary Tables S6–S8).

The phylogenetic tree topologies of the three genes were highly similar, each comprising five distinct clades (Clades I–V) (Fig. 3B). Clades within the same group, such as Clade I across the three gene trees, largely consisted of the same species (Fig. 3B), indicating that the *hsdR*, *hsdM*, and *hsdS* genes follow similar evolutionary trajectories. Among the five clade groups, Clade III contained the fewest phyla (seven), while Clade I encompassed the most phyla (nine) (Fig. 3B). For each clade, the gene trees displayed topologies somewhat similar to the species tree, suggesting that RMS is vertically transmitted within these species during evolution.

Desulfobacterota occupied a prominent position in the gene trees, often serving as the ancestral node for other phyla, such as Campylobacterota, Proteobacteria, and Bacteroidota (Fig. 3B). This suggests that RMS may have initially transferred from Desulfobacterota to other phyla, followed by vertical transmission within those phyla. Cyanobacteria also exhibited a similar pattern (Fig. 3B), indicating that Cyanobacteria could have been a significant donor of RMS in certain phyla. We hypothesize that the evolution of RMS involves a long and complex process (Fig. 3D). Initially, ancestral genes undergo duplication, followed by HGT to other species. As these species continue to evolve, the genes are either vertically inherited or lost. Since loose genomic loci cannot guarantee stable inheritance, the linkage between the *hsdR*, *hsdM*, and *hsdS* genes gradually becomes tighter over time, ultimately giving rise to the current RMS (Fig. 3D). While this model fits current evidence, future research should employ experimental evolution to test the hypothesis.

ΔRMS strain lacks flagellum and exhibits a reduced virulence phenotype

To explore the functional roles of RMS after its stable inheritance, we constructed the ΔRMS and ΔRMS/pRMS strains of *A. veronii* C4. These strains, along with the WT strain, were examined using TEM (Fig. 4A). The three strains exhibited similar cell sizes with rounded ends. In the WT strain, a single polar flagellum was clearly visible at one end, surrounded by short and dense pili. Detached flagella were also observed in the surrounding environment. In contrast, the ΔRMS strain lacked a polar flagellum and only contained pili, suggesting that RMS plays a key role in regulating flagellar assembly. Notably, the ΔRMS/pRMS strain still lacked a polar flagellum, indicating that RMS may not directly regulate flagellar assembly.

Flagellum is crucial for bacterial motility, and its absence may reduce bacterial pathogenicity. To test this hypothesis, healthy mice were infected with either the WT, ΔRMS, or ΔRMS/pRMS strain (Fig. 4B). Compared to the WT strain, the ΔRMS strain, which lacked the polar flagellum, showed a reduced ability to colonize various organs of mice, particularly the kidney (Fig. 4C). The ΔRMS/pRMS strain's colonization ability partially recovered, although it remained lower than that of the WT strain. Histopathological analysis revealed severe renal toxicity in mice infected with the WT strain, including enlarged renal tubules, swollen glomeruli, cellular necrosis, nuclear pyknosis, and interstitial congestion (Fig. 4D). Similar lesions were observed in the spleen

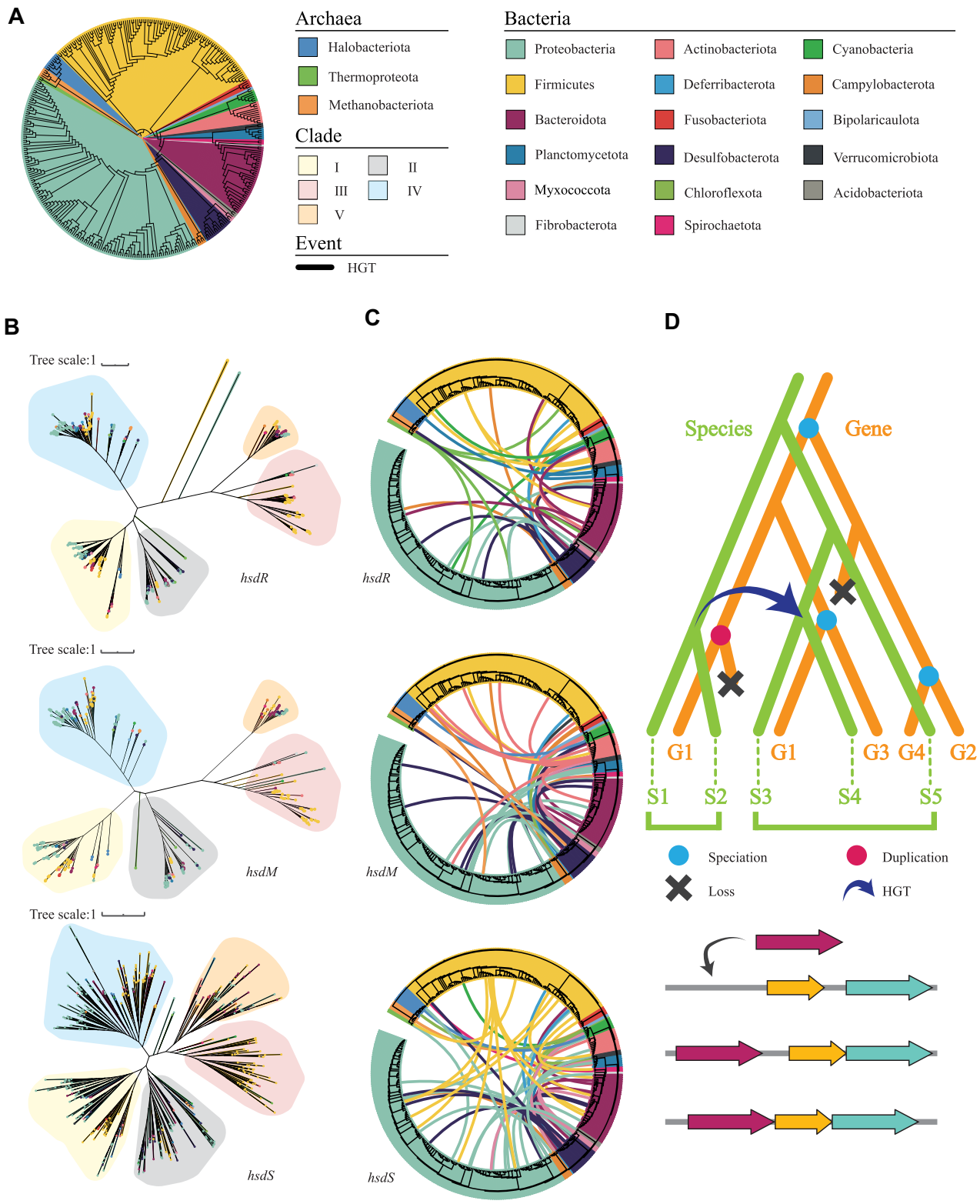


Figure 3. Phylogenetic analysis of RMS. (**A**) Phylogenetic tree of species containing the canonical Type I RM system (RMS), with different colors representing distinct phyla. (**B**) Phylogenetic trees of *hsdR*, *hsdM*, and *hsdS* genes, with different clades highlighted by varying background colors. Colored dots on tree branches indicate phylum affiliation, following the same color scheme as shown in panel (A). (**C**) HGT pathways of *hsdR*, *hsdM*, and *hsdS* genes. The outer circle represents the species tree from panel (A), while the inner curve indicates the HGT path. The color of curve corresponds to the donor species involved in each HGT event. (**D**) Evolutionary model of RMS development. Key stages: (i) duplication of ancestral RMS components; (ii) horizontal transfer to divergent species; (iii) loss of intervening genes; and (iv) stabilization into tightly linked RMS.

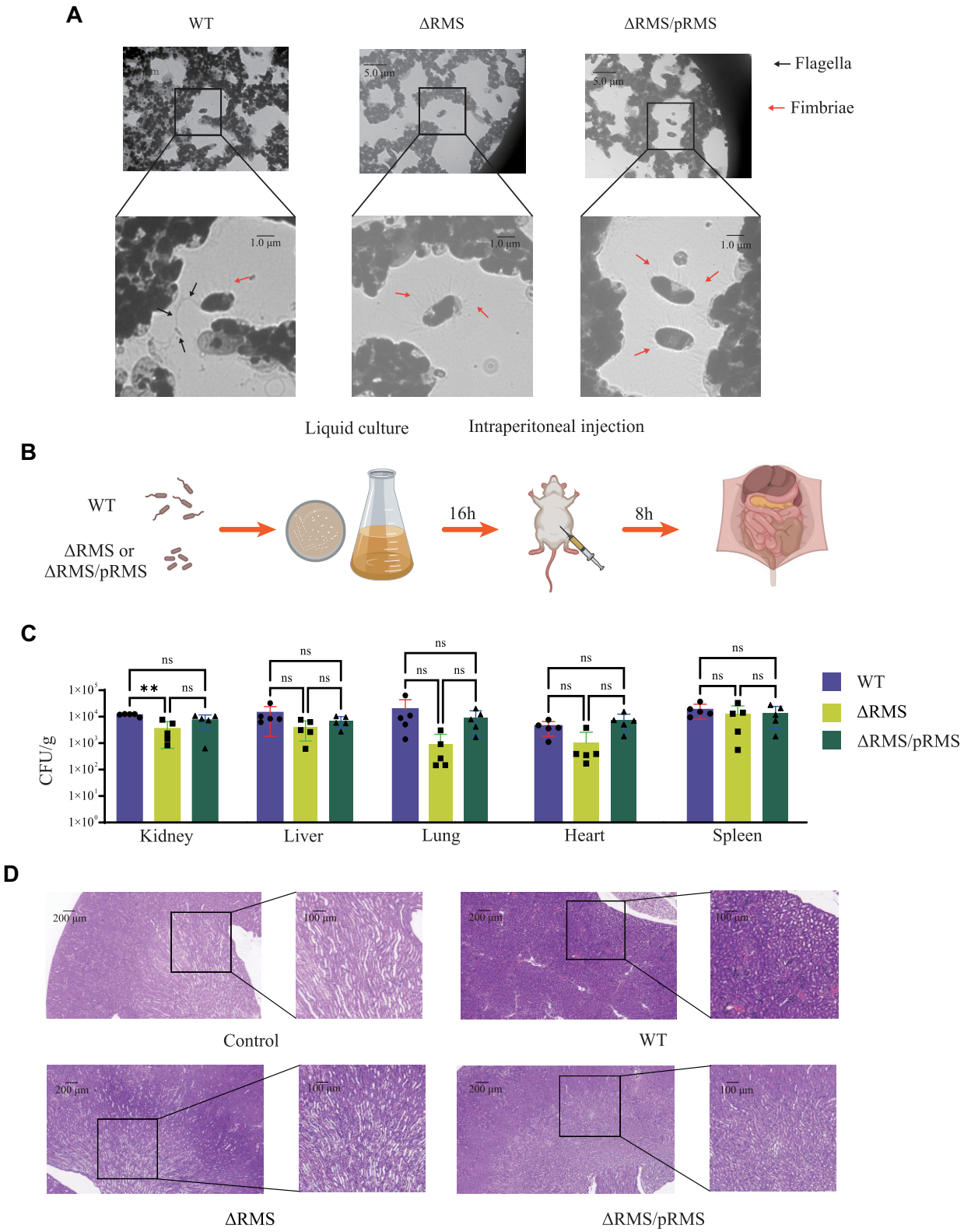


Figure 4. Impact of RMS on bacterial pathogenicity. (**A**) TEM images of *A. veronii* WT, RMS knockout (ΔRMS), and complemented (ΔRMS/pRMS) strains. Compared to the WT strain, the ΔRMS and ΔRMS/pRMS strains lack the polar flagellum. (**B**) Schematic illustration of mouse infection experiment. Mice were intraperitoneally injected with the WT, ΔRMS, or ΔRMS/pRMS strain at a dose of 10^5 CFU/g of body weight. (**C**) Colonization ability in different tissues of mice after challenged by the WT, ΔRMS, or ΔRMS/pRMS strain. Data are shown as mean ± SD, with the error bar indicates the standard deviation of five independent experiments. The significance of differences in colony counts among different strains is determined by one-way ANOVA test. ns indicates no significant difference and ** denotes $P < 0.01$. (**D**) Microscopic images of kidney tissues stained with H&E pathology slices. The control group shows the normal tissue morphology.

and liver, with inflammatory responses in the spleen and focal rupture of liver cells accompanied by nuclear aggregation (Supplementary Fig. S5). In contrast, infection with the ΔRMS strain did not induce obvious pathological changes. Notably, the ΔRMS/pRMS strain also caused renal tubule swelling, splenic inflammatory, and hepatocyte damage, although its virulence was still lower than that of the WT strain. The results indicated that the knockout of RMS substantially reduces the colonization ability and virulence of *A. veronii* C4.

RMS modulates the expression of downstream genes by methylating transcription factor-encoding genes

To investigate the molecular mechanisms by which RMS regulates bacterial pathogenicity, we sequenced the transcriptomes and methylomes of both WT and ΔRMS strains. Compared to the WT strain, 167 genes were differentially expressed in the ΔRMS strain, which were significantly enriched in pathways associated with flagellar assembly, bacterial chemotaxis, and more (Fig. 5A). Notably, these DEGs tended to cluster within the genome, forming gene clusters that encode such as the proteins involved in flagellar assembly and chemotaxis (Fig. 5B). This finding suggests that DEGs may be co-regulated by certain TFs.

In the WT strain, seven methylated motifs were identified (underlined = methylated bases), including three reverse-complementary pairs 5′-GTANNNNNNNNCTTC-3′↔5′-GAAGNNNNNNNNTAC-3′, 5′-CCAYNNNNNCTG-3′↔5′-CAGNNNNNRTGG-5′, 5′-TCAHNNNNNRTGGC-3′↔5′-GCCAYNNNNNDTGA-3′, and 5′-GATC-3′. In contrast, the ΔRMS strain only exhibited five 5′-CCAYNNNNNCTG-3′↔5′-CAGNNNNNRTGG-5′, 5′-TCAHNNNNNRTGGC-3′↔5′-GCCAYNNNNNDTGA-3′, and 5′-GATC-3′. The motif pair 5′-GTANNNNNNNNCTTC-3′↔5′-GAAGNNNNNNNNTAC-3′, which was absent in the ΔRMS strain, was the RMS target sites (Fig. 5C). A total of 386 sequences matching the RMS target sites were dispersed throughout the genome, with 330 of these sequences located within the bodies of 316 protein-coding genes. Functional enrichment analysis revealed these genes were significantly associated with amino acid metabolism and transmembrane transport processes (Supplementary Fig. S6).

Among the 316 analyzed genes, 16 encoded TFs, 9 of which (Supplementary Table S9) exhibited measurable regulatory activity. These nine TFs collectively controlled the expression of 89 genes (Supplementary Table S10) that were significantly downregulated in the ΔRMS strain. The DEGs included those encoding flagellar components (such as *flgO*, *flgP*, *fliC*, and *fliD*) and chemotaxis proteins (like *cheV*, *cheR*, and *mcp*). Through RT-qPCR validation (Fig. 6A), we confirmed reduced expression of both the TF-encoding genes (*rpoN*, *rpoS*, and *flrA*) and their respective targets: RpoN-regulated (*flgO* and *flgP*), RpoS-regulated (*fliC* and *fliD*), and Flra-regulated (*cheV* and *cheR*) genes. All examined genes except *rpoN* showed significant downregulation, with particularly strong effects observed for *rpoS*, *flgP*, *fliC*, and *fliD* (Fig. 6A). Notably, the nine TF-encoding genes contained both RMS target sites and a highly conserved motif 5′-TBVBCSABCTRHKCMABCTGCTGGAKCHG-3′ (*E*-value = 5.30×10^{-10}; Supplementary Fig. S7). However, the 89 downstream-regulated DEGs completely lacked RMS target sites, indicating that RMS indirectly regulates downstream genes, including those involved in flagellar assembly and chemotaxis, ultimately influencing bacterial pathogenicity.

Discussion

RM systems serve as key DNA defense mechanisms in prokaryotes, playing a crucial role in protecting hosts against foreign genetic elements, such as phages. These systems are traditionally classified into four types (I–IV) based on their structural and functional characteristics [8]. Although Types II and III RM systems have been extensively studied, significant knowledge gaps persist regarding Type I RM system despite its widespread occurrence in prokaryotes. Specifically, three key aspects remain poorly understood: (i) its distribution patterns across prokaryotes, (ii) its evolutionary origin and long-term evolutionary trajectory, and (iii) its functional role following stable inheritance as well as underlying molecular mechanism.

Through a systematic analysis of 4273 prokaryotic genomes, we found that only 64.6% (*hsdR*), 75.5% (*hsdM*), and 60.2% (*hsdS*) of species carry genes encoding Type I RM systems, suggesting that at least a quarter of species may rely on alternative defense mechanisms, such as Type II/III RM systems or CRISPR–Cas systems, to protect their genomic integrity [45]. Further analysis revealed that four main gene arrangements of Type I RM systems exist in prokaryotes. Among them, RMS and RSM, consisting of tightly linked *hsdR*, *hsdM*, and *hsdS* genes, were more frequently observed in bacteria compared to R-MS and R-SM, which have gene insertions. While archaea showed opposite trends with R-MS/R-SM predominance. This finding is in strong alignment with evolutionary theories. Under evolutionary pressures, early Type I RM systems, constructed from pre-existing components, evolve into "simpler" systems through domain fusion or deletion [46]. This suggests that R-MS may represent an earlier form of RMS, while R-SM may be an early version of RSM. Over time, genes inserted between *hsdR* and *hsdMS*/*hsdSM* are progressively lost, eventually leading to the formation of the more streamlined RMS and RSM. Furthermore, the close proximity of *hsdR* and *hsdM*/*hsdS* on the genome reduces the likelihood of random imbalances between these genes, minimizing the risk of self-restriction, which could trigger the host's SOS response [47]. This may explain why the host tends to favor RMS or RSM as the final form of Type I RM systems.

The comparative analysis at the phylum level revealed evolutionary dynamics. While the phylum Cyanobacteria exhibited abundant copies of individual *hsdR*, *hsdM*, and *hsdS* genes, the complete Type I RM systems showed remarkably low occurrence frequencies. Given that the emergence of Type I RM systems is thought to be based on pre-existing components [48], the observed pattern in Cyanobacteria likely reflects an early stage in the evolutionary process of Type I RM systems. In contrast, the phylum Desulfobacterota exhibited distinctly different evolutionary characteristics: it not only harbored a significantly higher abundance of complete Type I RM systems, but also maintained elevated copies of individual *hsd* genes. Particularly noteworthy was the observation that the transitional configuration R-SM occurred at substantially higher frequencies than the fully developed architecture RSM

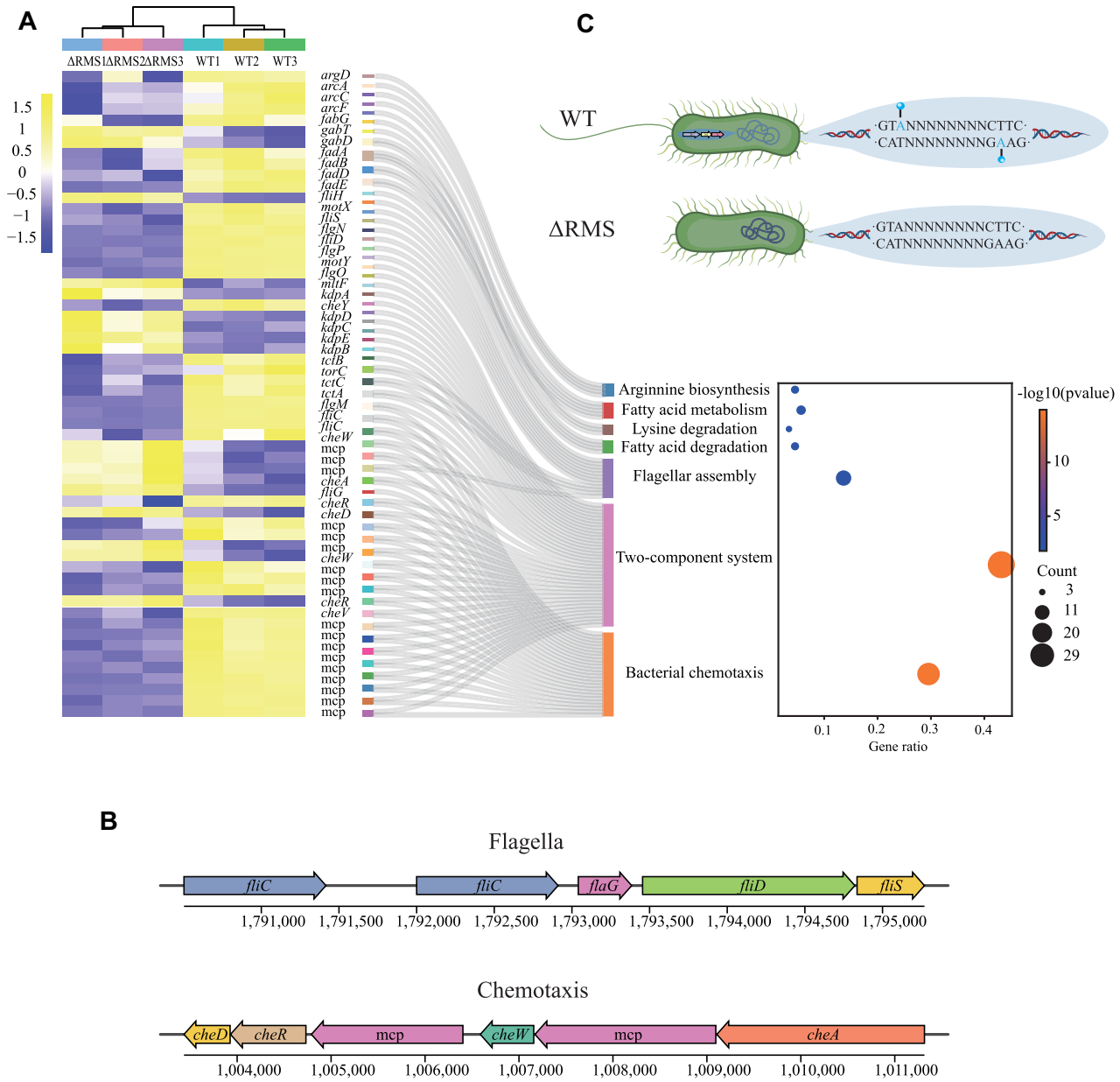


Figure 5. Pathways regulated by RMS. (**A**) KEGG functional enrichment analysis of DEGs. The heatmap displays their expression levels in *A. veronii* WT and RMS knockout (ΔRMS) strains, with colors ranging from purple (low expression) to yellow (high expression). Sankey and bubble plots illustrate the pathways enriched by DEGs. (**B**) Gene clusters comprising partial DEGs. (**C**) Motif identified and methylated by RMS. Following RMS knockout, the corresponding sequences in the ΔRMS strain lose methylation modifications.

within this phylum. This unique distribution pattern strongly suggests that Type I RM systems in Desulfobacterota are currently undergoing rapid evolution. We propose the following evolutionary trajectory: First, individual *hsd* genes progressively assemble into novel Type I RM systems. Concurrently, the more primitive R-SM undergo streamlining processes (e.g. gene loss) to evolve into the more stable RSM. This differential evolutionary pattern likely reflects strong environmental adaptation pressures.

To elucidate the key factors shaping the distribution patterns of Type I RM systems, we conducted a comparative genomic analysis using *A. veronii* as a model organism, examining strains isolated from diverse ecological niches. Our findings revealed that aquatic isolates exhibited a significantly higher number of Type I RM systems and prophages compared to strains from other environments. This could be due to the favorable conditions for phage survival in aquatic environments, which enable strains to integrate more phage sequences into its genome, resulting in a higher number of prophages. Moreover, strains may need to evolve a greater number of Type I RM systems as a defense mechanism against phages [49]. Notably, our study also revealed distinct geographical distribution patterns: Chinese and Japanese isolates were enriched with Type I RM systems, while Thailand isolates completely lacked them. This geographical correlation is further supported by studies on *Helicobacter pylori*, where RM system diversity correlates with strain origins across continents [50]. Some RM systems

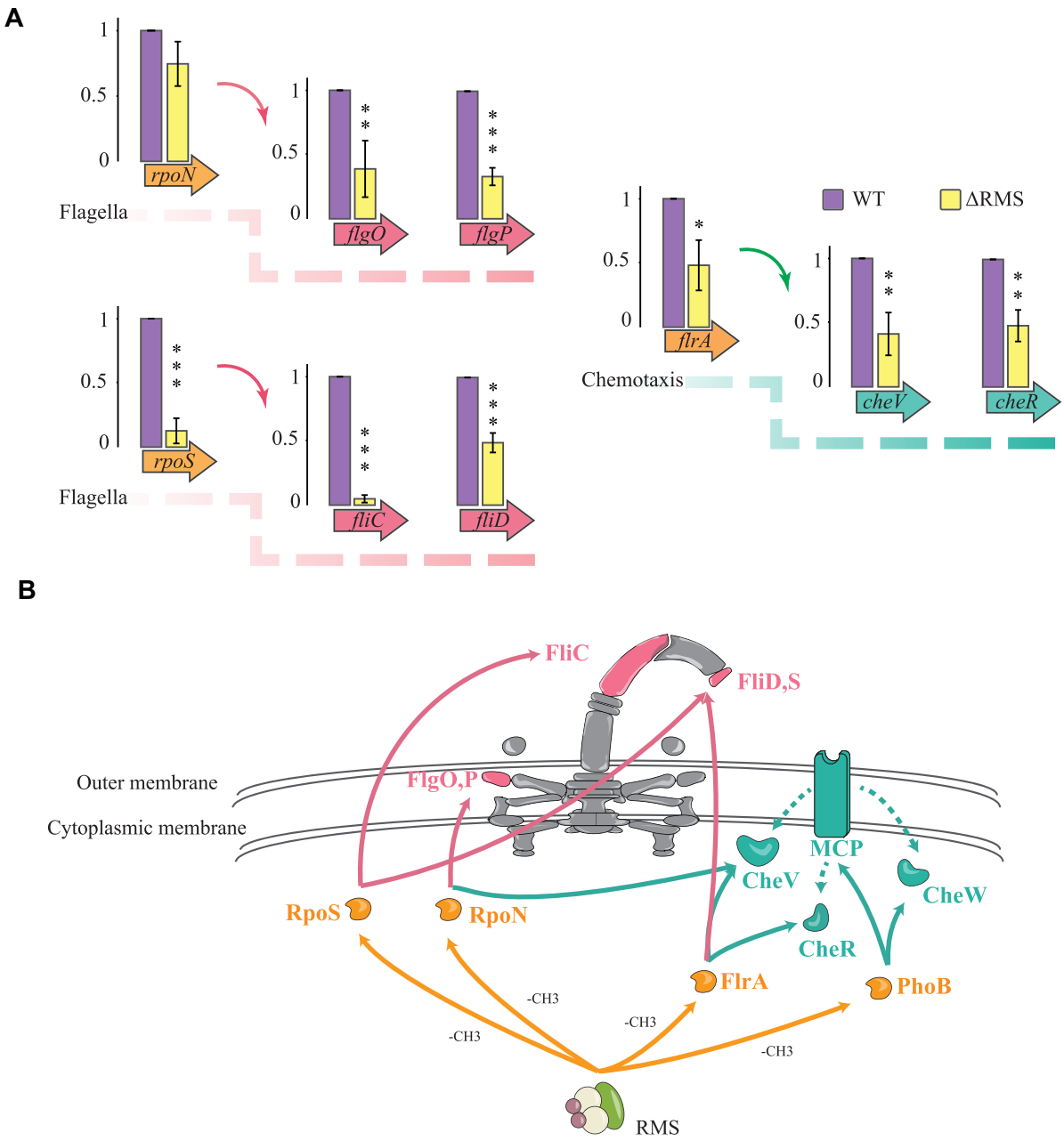


Figure 6. Regulation mechanisms of flagellar assembly and chemotaxis by RMS. (**A**) RT-qPCR validation of DEGs. The orange arrow indicates the transcription factor (TF), while the green and red arrows represent their target genes, encoding chemotaxis proteins and flagellar proteins, respectively. Data are shown as mean ± SD, with the error bar indicates the standard deviation of three independent experiments. The significance of differences in gene expression levels between *A. veronii* WT and ΔRMS strains is assessed using one-way ANOVA test. $*P < 0.05$, $**P < 0.01$, and $***P < 0.001$. (**B**) Schematic diagram illustrating the hypothetical regulatory model by which RMS regulates flagellar assembly and chemotaxis. RMS initially regulates TFs (indicated in orange) through DNA methylation, which in turn control the expression of flagellar proteins (indicated in red) and chemotaxis proteins.

may have been acquired during early human migrations out of Africa, while others are independently obtained by geographically isolated populations later [51]. These findings suggest that defense systems, including Type I RM systems, play crucial roles in long-term bacterial evolution and host adaptation.

The evolutionary dynamics of RM systems have long been recognized to involve HGT [52, 53]. However, significant gaps remain in our understanding of HGT in Type I RM systems. By performing phylogenetic reconciliation between species trees and individual gene trees (*hsdR*, *hsdM*, and *hsdS*), we have obtained robust evidence for widespread HGT in the canonical Type I RM system (RMS). HGT serves as a primary evolutionary force driving the remarkable diversity of *hsd* gene arrangements, which in turn generates the various configurations of Type I RM systems. Particularly noteworthy is our demonstration of extensive HGT occurring at multiple taxonomic levels—both within and between phyla. This broad-scale mobility strongly supports the hypothesis that RMS likely serves as a communal resource, contributing to prokaryotic pan-immunity [15]. Phylogenetic classification of *hsd* genes identified five distinct clades with contrasting

evolutionary patterns. Clade III, containing the fewest phyla, appears to represent a more recently evolved group with limited HGT. In contrast, Clade I exhibited the broadest phylum-level distribution, suggesting both an ancient evolutionary origin and sustained cross-phylum dissemination through extensive HGT. These findings align well with current understanding of immune system evolution in prokaryotes. Immune systems are energetically costly and are retained and vertically transmitted only under selective pressure [54, 55]. When selective pressure is low, however, immune systems are often transferred via mobile genetic elements (MGEs), which keep them in a state of constant flux [15, 56]. Although the current study offers a systematic characterization of *hsd* gene evolutionary dynamics at the prokaryotic scale, subsequent research should prioritize selected model organisms to dissect strain-specific *hsd* genomic rearrangements. By employing integrated approaches, these investigations will elucidate the synergistic effects of MGEs, recombination events, and niche-specific selection pressures in orchestrating the adaptive evolution of RMS.

The evolutionary persistence of RMS suggests that it may confer dual selective advantages: maintaining genome integrity through RM while epigenetically regulating critical biological processes. To dissect the function of RMS after stable inheritance, we constructed the ΔRMS and ΔRMS/pRMS strains for comprehensive phenotypic characterization. TEM anlysis revealed that the WT strain exhibited an intact flagellum, whereas the ΔRMS strain completely lost its flagellum (retaining only pili), indicating that RMS plays a crucial regulatory role in flagellar assembly. Notably, the ΔRMS/pRMS strain failed to restore flagellar formation, suggesting that RMS exerts its function through indirect regulatory mechanisms rather than directly controlling flagellar gene expression. These findings are strongly supported by *in vivo* infection assays: Compared to the WT strain, the ΔRMS strain displayed reduced colonization in murine kidneys, spleens, and livers, and caused no evident histopathological damage. Although complementation restored colonization and virulence, it still did not reach the levels observed in the WT strain. This phenotypic pattern also closely parallels findings in *S. pyogenes*, where deletion of RSM similarly attenuates virulence with only partial restoration in the complemented strain [17]. Such cross-species consistency implies that Type I RM systems likely govern bacterial pathogenicity via indirect mechanisms. Furthermore, the results also provide new insights into the genetic organization requirements of Type I RM systems. Although the genomic arrangement of *hsdR*, *hsdM*, and *hsdS* varies among different systems (e.g. RSM and RMS), these differences do not appear to affect core functionality. In contrast, the transcriptional orientation of *hsd* genes is critical for the functionality of Type I RM systems. When *hsdS* and *hsdM* are encoded on opposite strands, one of the genes may fail to be expressed due to the absence of a promoter [57].

Integrative multi-omics analyses further substantiate our finding that RMS played no direct role in regulating bacterial pathogenicity. Only 4.8% (8/167) (Supplementary Table S11) of the DEGs detected upon RMS deletion possessed the RMS target sites 5′-GTANNNNNNNNCTTC-3′↔5′-GAAGNNNNNNNNTAC-3′. They were specifically recognized by HsdS, which contains two structurally independent target recognition domains (TRDs) connected by α-helical linkers. Each TRD exhibits distinct sequence specificity: one recognizes the sequence 5′-GTA-3′↔5′-TAC-3′ and the other binds to the sequence 5′-CTTC-3′↔5′-GAAG-3′. The modular architecture of HsdS enables remarkable phase variation through TRD recombination, generating diverse sequence specificities across strains and species. A compelling example is seen in *H. pylori*. P12 strain (HsdS1 with TRDs c + a) recognizes 5′-GAA N8 TAG-3′↔5′-CTA N8 TTC-3′. F16 strain (HsdS2 with TRDs h + d) recognizes 5′-CAA N7 RTAB-3′↔5′-VTAY N7 TTG-3′. Recombinant variant (HsdS3 with TRDs a + h) recognizes 5′-CAA N8 TAG-3′↔5′-CTA N8 TTG-3′ [5]. This TRD shuffling mechanism represents a powerful evolutionary driver that generates methylome diversity and facilitates bacterial adaptation. In the future, we will systematically characterize HsdS domain rearrangements, with the goal of reconstructing the evolutionary history of Type I RM systems.

Genome-wide localization analysis further revealed that the RMS target sites were preferentially enriched within gene bodies rather than promoter regions, suggesting a non-canonical regulatory mode that differs from traditional TF-mediated promoter regulation. This observation aligns with emerging paradigms of intragenic DNA methylation in eukaryotes, where gene body methylation effects are explained by two prevailing hypotheses. One is derived from the anticorrelation between H2A.Z histone variant distribution and DNA methylation. This hypothesis proposes that gene body methylation maintains transcriptional continuity by inhibiting H2A.Z deposition [58]. The other posits that gene body methylation enhances transcriptional efficiency by suppressing aberrant initiation from alternative promoters and/or antisense transcription [58–60]. We speculate that in prokaryotes, RMS-mediated gene body methylation may exert analogous effects on gene expression. For instance, through modulation of nucleoid-associated protein (NAP)–DNA interactions, wherein methylation sterically hinders NAP binding and consequently alters local DNA topology and RNA polymerase processivity. Alternatively, via regulation of functional genomic elements, whereby methylation suppresses spurious transcription initiation. This process may involve auxiliary factors that potentially recognize and bind to conserved motifs within TFs, e.g. the motif 5′-TBVBCSABCTRHKCMABCTGCTGGAKCHG-3′, thereby cooperating with RMS to jointly modulate TFs. These putative mechanisms, however, require systematic experimental validation.

Genomic analysis identified 16 TF-encoding genes containing RMS target sites, among which nine TFs were found to regulate 89 DEGs involved in critical pathogenicity-associated processes, including flagellar assembly and bacterial chemotaxis. Supported by RT-qPCR validation, we postulate a potential cascade regulation model (Fig. 6B): RMS may fine-tune the expression of these TF-encoding genes through DNA methylation, while these TFs in turn act as regulatory hubs modulating downstream genes, ultimately governing bacterial pathogenicity. This hierarchical regulation exhibits remarkable signal amplification; even subtle changes in TF expression can cause substantial fluctuations in downstream targets [61, 62]. However, we emphasize this represents one possible mechanism among the organism's diverse regulatory repertoire. In conclusion, our study offers new insights into how RMS achieves stable genomic inheritance—through horizontal transfer followed by functional integration into pathogenicity networks, where it could serve as high-level epigenetic modulator rather than direct transcriptional controller.

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Author contributions: J.M. was involved in conceptualization, methodology, investigation, formal analysis, visualization, and writing—original draft. X.J. was involved in methodology, investigation, and writing—original draft. H.B. was involved in formal analysis, visualization, and data curation. J.L. was involved in investigation. X.M. was involved in investigation. X.C. was involved in formal analysis. Y.T. was involved in formal analysis. Z.L. was involved in conceptualization, writing—review and editing, and supervision. H.L. was involved in conceptualization, writing—original draft, writing—review and editing, and supervision.

Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

RNA and DNA methylation sequencing data are available at Sequence Read Archive in National Center for Biotechnology under accession numbers PRJNA1200627 and PRJNA1010177.

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